



PREDICTION OF APPENDICITIS DIAGNOSIS USING MACHINE LEARNING WITH CLINICAL AND LABORATORY DATA

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Abstract

Appendicitis is a common pediatric emergency where accurate and timely diagnosis is often difficult due to nonspecific symptoms. This study uses the Regensburg Pediatric Appendicitis dataset, which includes data from 782 patients along with 53 clinical, laboratory, and ultrasound variables, to create predictive models. Through statistical analysis, data visualization, and machine learning, we examine how key clinical and laboratory factors influence diagnosis.

Our results identify important predictors such as inflammatory markers and ultrasound results, and machine learning models, especially ensemble models, are good at classifying appendicitis. Explainable AI techniques provide insight into the individual contribution of every feature to predictions, which enhances model interpretability for clinical practice.

The findings suggest that machine learning can help reduce misdiagnosis, prevent unnecessary surgeries, and enable quicker treatment decisions. An interactive dashboard and a user-friendly application turn these insights into practical tools for clinicians.

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Introduction

Pediatric appendicitis is a common surgical emergency requiring timely and accurate diagnosis to prevent serious complications like perforation. However, diagnosis is often difficult due to nonspecific symptoms and overlapping clinical signs, especially in younger children. Traditional methods clinical evaluation, lab tests, and imaging can be variable and subjective, leading to misdiagnosis or unnecessary surgeries. Machine learning can enhance diagnosis by integrating diverse data sources to provide rapid, objective, and accurate risk assessments. Using the comprehensive Regensburg Pediatric Appendicitis dataset, ML models have the potential to improve clinical decision-making and patient outcomes by identifying patterns beyond conventional approaches.

Description of the question

This study focuses on pediatric diagnosis of appendicitis to investigate clinical, lab, and ultrasound variables impacting proper detection of the disease. With nonspecific presentation, overlapping clinical signs, and possible unnecessary surgery or delayed treatment, higher diagnostic accuracy is important for expediting and enhancing patient care.

We focus on two main objectives:

➤ Analyze Predictors of Appendicitis

We evaluate how demographic, clinical, laboratory, and ultrasound variables contribute to distinguishing true appendicitis cases from other conditions, uncovering diagnostic patterns and irregularities.

➤ Build Predictive and Explainable Models

We apply statistical and machine learning models to predict appendicitis. Using feature importance and explainable AI tools, we quantify how key factors drive diagnostic outcomes and integrate these insights into an interactive dashboard and user-friendly application.

Together, these objectives aim to provide actionable insights for clinicians. By combining diverse medical data with explainable machine learning, this study supports evidence-based strategies to reduce misdiagnosis, avoid unnecessary surgeries, and improve pediatric healthcare outcomes.

Description of the data set

The “**Regensburg Pediatric Appendicitis**” dataset, sourced from the UCI Machine Learning Repository, presents a comprehensive collection of clinical, laboratory, and ultrasound findings relevant to the diagnosis of appendicitis in children. The dataset contains **782 rows (patients)** and **53 columns (variables)**. Each row represents a unique pediatric patient admitted with abdominal pain, and the columns capture their demographic information, laboratory test results, physical examination findings, clinical scoring systems, and expert-assessed ultrasound observations. The primary target variable is the **confirmed diagnosis of appendicitis** (yes/no).

However, it has certain limitations such as,

Single-Center Data: The data is collected from one hospital (Children's Hospital St. Hedwig, Regensburg, Germany), which may limit generalizability of findings to larger populations.

Missing and Imbalanced Data: There are missing values for certain variables, and class imbalance between appendicitis and non-appendicitis may affect model performance.

Limited Scope of Outcomes: While additional labels such as management (surgical vs. conservative) and severity (complicated vs. uncomplicated) are available, this study focuses primarily on diagnosis.

Category	Example Variables	Type
Demographics	Age, Sex, Height, Weight, BMI	Continuous, Categorical
Hospital Information	Length_of_Stay, Management, Severity	Categorical, Integer
Diagnosis	Diagnosis (confirmed), Diagnosis_Presumptive	Binary
Clinical Scores	Alvarado_Score, Pediatric_Appendicitis_Score	Integer
Symptoms & Signs	Migratory_Pain, Nausea, Loss_of_Appetite, Lower_Right_Abd_Pain, Rebound Tenderness	Binary/Categorical
Vital Signs	Body_Temperature	Continuous
Laboratory Tests	WBC_Count, Neutrophil_Percentage, Hemoglobin, RBC_Count, CRP, Urine findings	Continuous/Categorical
Ultrasound Findings	Appendix_on_US, Appendix_Diameter, Free_Fluids, Target_Sign, Perforation, Abscess	Binary/Categorical

Other Abdominal Findings	Lymph_Nodes, Bowel_Wall_Thickening, Ileus, Coprostasis, Enteritis, Gynecological_Findings	Binary/Categorical
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Table 1: Description of the variables

Data Preprocessing

In the data pre-processing phase, the appendicitis dataset of 782 paediatric patients and 58 variables was cleaned and prepared for analysis. High-missing columns (>60%) were removed because the proportion of missing data was too large to allow for reliable imputation, and retaining these variables could introduce noise or bias into the analysis without adding meaningful value. It was noticed that, except for one variable ('Segmented_Neutrophils'), all of the other variables that were removed related to Ultrasound data.

Apart from missing values, we proceed to remove several other variables to maintain the reliability of the analysis and the predictive model.

Variable Removed	Reason
Length_of_Stay	This is an outcome-related variable reflecting hospital resource use and patient recovery duration, which occurs after diagnosis and treatment decisions. It is not a predictor but a downstream consequence, so including it would cause data leakage and bias in diagnostic modelling.
US_Performed	This variable indicates whether ultrasound imaging was done, not the diagnostic content itself. Whether the US is performed depends on clinical judgment after initial suspicion and is not a direct predictive feature for diagnosis. Including it could introduce procedural bias rather than information valuable to the initial diagnosis.
Diagnosis_Presumptive	This variable represents a preliminary assumption of diagnosis before definitive confirmation. Since our task is to predict the definitive diagnosis, including presumptive diagnosis may create circular reasoning, artificially inflating predictive accuracy.
Management	Management decisions depend on diagnostic outcomes and clinician strategies post-diagnosis. This variable is subsequent to diagnosis, so including it as a predictor can introduce post-outcome bias.
Severity	Severity is a further subclassification of confirmed appendicitis cases and thus is downstream from diagnosis. It is relevant to prognosis or management but not appropriate as an input feature for predicting initial diagnosis.
US_Number	The variable "US_number" represents the number of ultrasound images or views acquired for each patient. This variable is a procedural or metadata feature reflecting how many ultrasound scans were performed, rather than any direct clinical or diagnostic information about the patient's condition.

Table 2: Removed Variables

Before imputing the missing values, the dataset was split into two sets, the training set and the testing set, such that 20% of the data will be in the testing set. After that, missing values were imputed for both sets separately, using the values from the training set.

Missing values in numerical features were imputed using either the median or the mean, depending on the skewness of the distribution, while categorical features were imputed with the mode. Features with skewed distributions were imputed using the median to minimize the influence of outliers, while normally distributed features were imputed using the mean to preserve statistical properties. Outliers were identified using the IQR method, and no duplicates were present. Variables were classified as continuous, integer, binary, or categorical. These steps ensured a clean, consistent, and analysis-ready dataset for modelling paediatric appendicitis diagnosis.

Descriptive Analysis

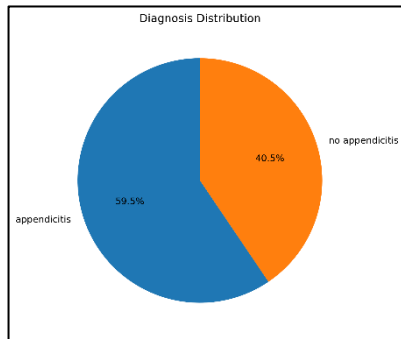


Figure 1: Diagnosis Distribution

The chart shows that most pediatric patients are diagnosed with **appendicitis**, while a smaller proportion are labeled as **no appendicitis**. This indicates that in this sample, appendicitis is the predominant outcome among children presenting with abdominal pain, highlighting the importance of accurate diagnosis and timely intervention.

The bar chart shows that more males are diagnosed with appendicitis than females, while the proportion of no appendicitis is slightly higher among females. This indicates a slightly higher prevalence of appendicitis in males.

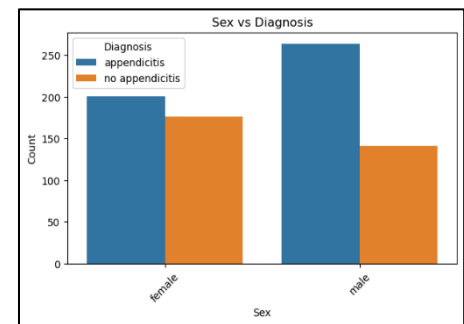


Figure 2: Sex vs diagnosis

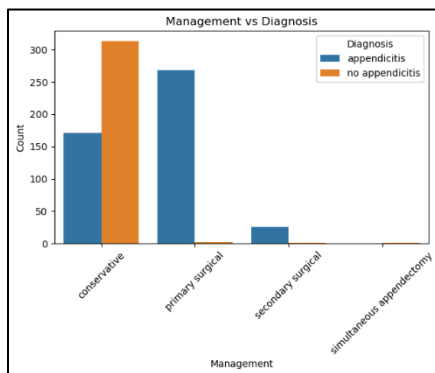


Figure 3: Management vs diagnosis

Patients managed conservatively mostly did not have appendicitis, while those who underwent primary or secondary surgical management were predominantly diagnosed with appendicitis. This suggests that surgical intervention is strongly associated with confirmed appendicitis.

The chart compares ultrasound detectability of the appendix with diagnosis. When the appendix is visible on ultrasound, most cases are confirmed as appendicitis. Non-visualized appendices mostly correspond to no appendicitis, showing ultrasound detectability as an important diagnostic feature.

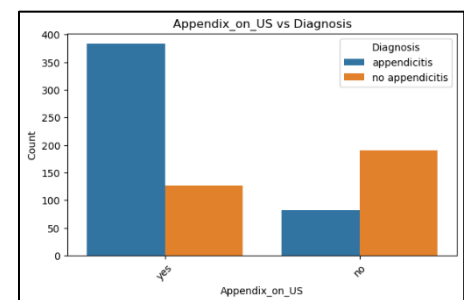


Figure 4: Appendix on US vs diagnosis

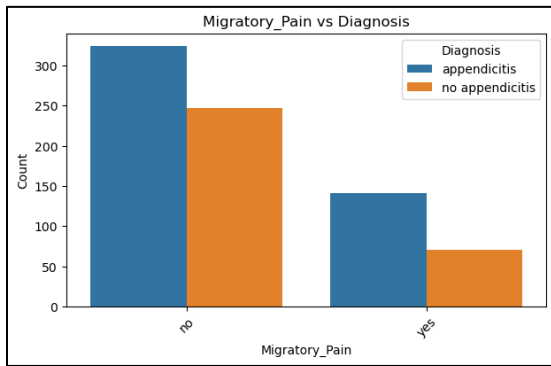


Figure 5: Migratory Pain Vs Diagnosis

This chart shows the relationship between right lower abdominal (RIF) pain and appendicitis diagnosis. Most patients report RIF pain, and among them, appendicitis is highly prevalent. Conversely, patients without RIF pain mostly do not have appendicitis. This indicates that right lower abdominal pain is a strong clinical indicator for appendicitis in this dataset.

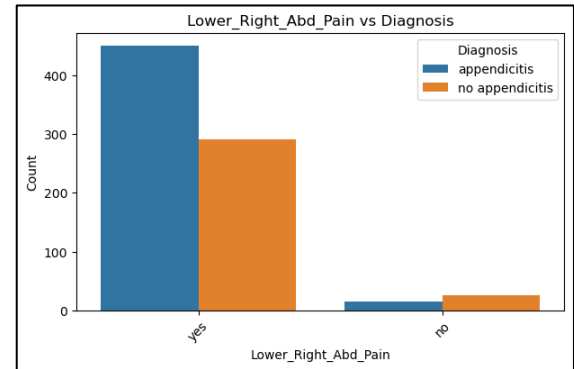


Figure 6: lowerRight Abd Pain vs vs diagnosis

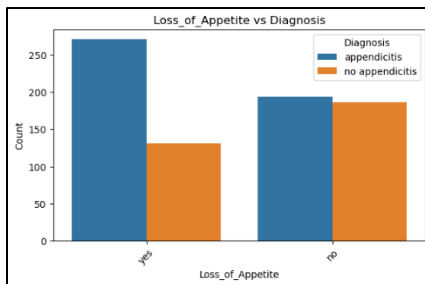


Figure 7: loss of appetite vs diagnosis

Loss of appetite is common in appendicitis patients, while non-appendicitis patients mostly reported normal appetite, making it a strong symptom.

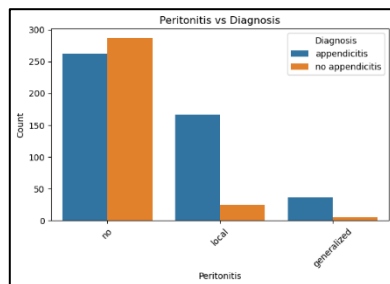


Figure 9: Peritonitis vs vs diagnosis

This chart shows the type of peritonitis (No, Local, Generalized) versus appendicitis diagnosis. Most patients without appendicitis have no peritonitis. Among patients, no peritonitis is the most common, followed by a smaller number with generalized peritonitis.

This bar chart compares the presence of free fluids in the abdomen with appendicitis diagnosis. Most patients without free fluids are roughly equally split between appendicitis and no appendicitis. In contrast, among patients with free fluids, the majority have appendicitis, while relatively few do not. This suggests that detecting free fluids is a strong sonographic indicator of appendicitis.

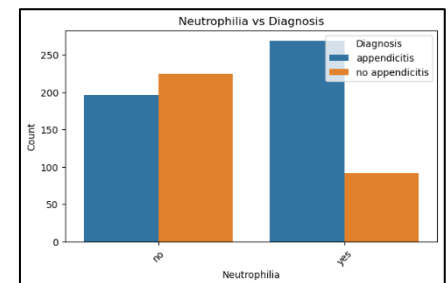


Figure 8: neutrophilia vs vs diagnosis

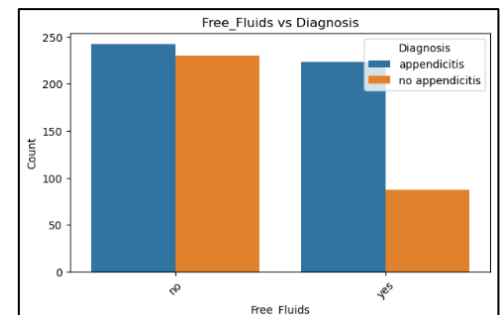


Figure 10: free fluids vs vs diagnosis

The heatmap shows strong positive correlations among growth measures (age–height $r = 0.85$, height–weight $r = 0.82$, weight–age $r = 0.76$) and between clinical scores (Pediatric Appendicitis and Alvarado $r = 0.83$). Inflammatory markers also align, with WBC–neutrophils ($r = 0.62$), CRP–length of stay ($r = 0.50$), and hemoglobin–RBC ($r = 0.43$) showing notable links. Moderate associations include Alvarado with WBC ($r = 0.62$), body temperature with WBC ($r = 0.47$), and neutrophils with Alvarado ($r = 0.67$). Most other variables, such as RDW, US Number, and RBC count, show negligible relationships, while length of stay only relates meaningfully to CRP. Negative correlations are seen for age with body temperature ($r = -0.29$), thrombocyte count with temperature ($r = -0.23$), and BMI with temperature ($r = -0.15$).

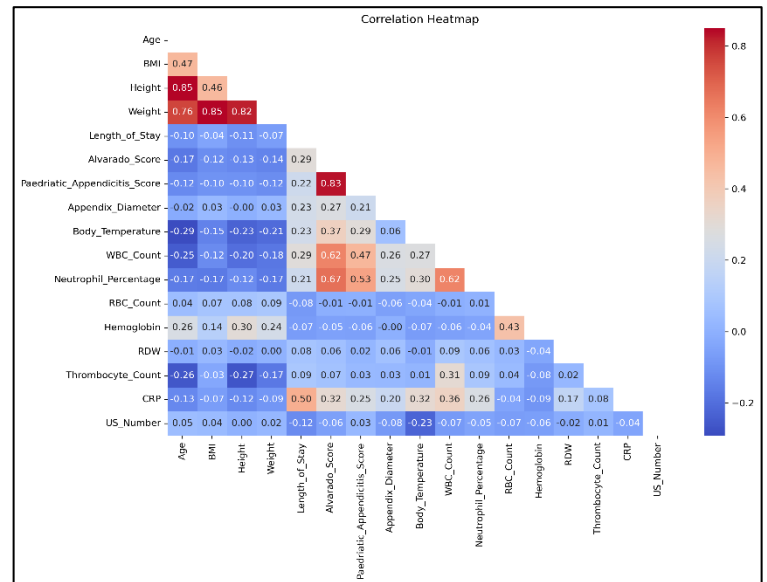


Figure 11: Correlation Matrix

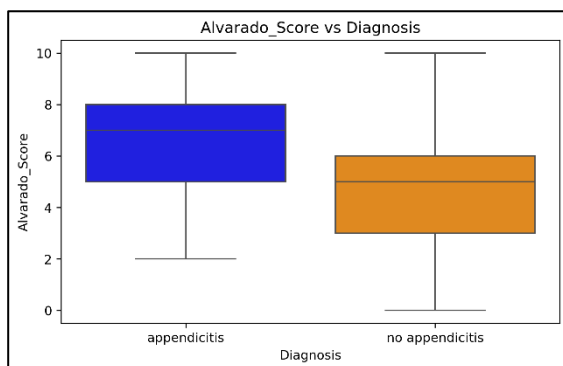


Figure 12: Alvarado_Score vs Diagnosis

The box plot "Alvarado_Score vs Diagnosis" shows the distribution of Alvarado scores for both appendicitis and non-appendicitis patients. The median score is higher and the interquartile range is narrower for appendicitis patients compared to non-appendicitis patients, which suggests that higher Alvarado scores increase the likelihood of appendicitis..

The box plot "Appendix_Diameter vs Diagnosis" graphs appendix diameter for both groups against one another. Patients with appendicitis have larger median and more spread-out range of appendix diameters, and the non-appendicitis patients have smaller diameters. This indicates that larger appendix diameter is related to appendicitis diagnosis.

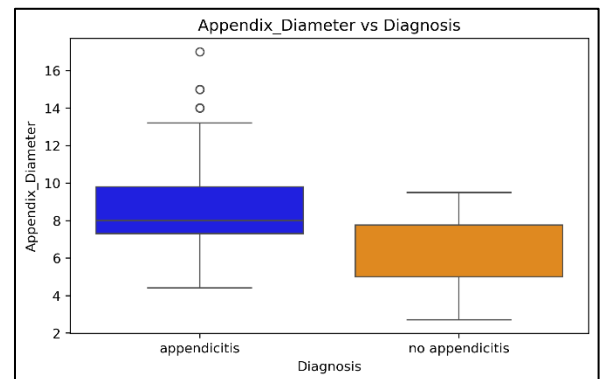


Figure 13: Appendix_Diameter vs Diagnosis

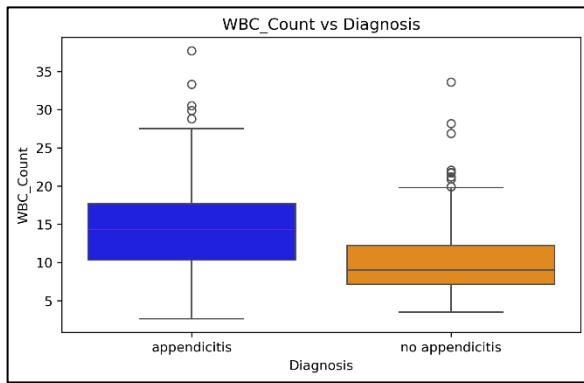


Figure 14: WBC_Count vs Diagnosis

inflammation. The box plot titled "*Thrombocyte_Count vs Diagnosis*" compares Thrombocyte count distributions between the groups. Both groups show approximately similar median values. This suggests the Thrombocyte count may have limited discriminative power between diagnoses.

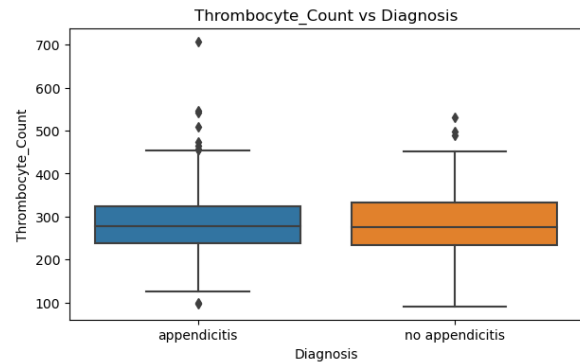


Figure 15: Thrombocyte count Vs Diagnosis

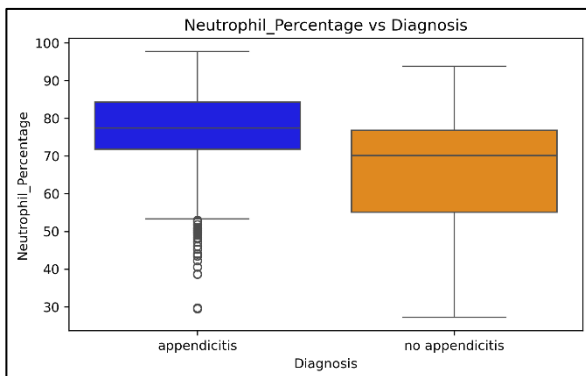


Figure 16: Neutrophil_Percentage vs Diagnosis

The box plot titled "*Neutrophil_Percentage vs Diagnosis*" illustrates the distribution of neutrophil percentages across both groups. Appendicitis patients show a higher median and tighter clustering around elevated values, while non-appendicitis patients display a slightly lower median and wider spread. This suggests elevated neutrophil percentage is strongly associated with appendicitis.

Statistical Tests

Pearson's Chi-squared test with Yates' continuity correction

```
data: table(Sex, Diagnosis)
X-squared = 9.5377, df = 1, p-value = 0.00213
```

Table 3: Chi-square test for association between Sex Vs Diagnosis

$\chi^2(1) = 9.54, p = .002$. This finding suggests that the distribution of diagnoses differed significantly between males and females.

Welch Two Sample t-test

```
data: Neutrophil_Percentage by Diagnosis
t = -8.0373, df = 494.49, p-value = 6.81e-15
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
95 percent confidence interval:
-10.703234 -6.498204
sample estimates:
mean in group 0 mean in group 1
66.33625 75.13697
```

Table 4: t-test for the difference in Neutrophil percentage between Diagnosis groups

the two diagnostic groups, $t = -8.04, p < .001$. Patients without diagnosis (66.54%) had a mean Neutrophil Percentage approximately 8.6 percentage points lower than patients with diagnosis (75.14%).

Welch Two Sample t-test

```
data: Appendix_Diameter by Diagnosis
t = -13.103, df = 611.46, p-value < 2.2e-16
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
95 percent confidence interval:
-2.141909 -1.583540
sample estimates:
mean in group 0 mean in group 1
6.701030 8.563754
```

Similarly, a significant difference was found in Appendix Diameter, $t = -13.10$, $p < .001$, with patients without diagnosis (6.70) showing a mean Appendix Diameter approximately 1.86 units smaller than those with diagnosis (8.56).

Table 5: *t*-test for the difference in Appendix diameter between Diagnosis groups

```

welch Two Sample t-test

data: WBC_Count by Diagnosis
t = -9.6871, df = 600.41, p-value < 2.2e-16
alternative hypothesis: true difference in means between group 0 and group 1 is not equal to 0
95 percent confidence interval:
 -4.564724 -3.025845
sample estimates:
mean in group 0 mean in group 1
 10.34129      14.13657

```

Table 6: *t*-test for the difference in WBC count between Diagnosis groups

```

Pearson's Chi-squared test with Yates' continuity correction

data: table(Neutrophilia, Diagnosis)
X-squared = 42.519, df = 1, p-value = 7e-11

```

Table 7: *Chi-square test for association between Neutrophilia Vs Diagnosis*
association between Neutrophilia and the presence of appendicitis

```

Pearson's Chi-squared test with Yates' continuity correction

data: table(Free_Fluids, Diagnosis)
X-squared = 26.635, df = 1, p-value = 2.458e-07

```

Table 8: *Chi-square test for association between Free Fluids Vs Diagnosis*

Finally, White Blood Cell (WBC) counts also differed significantly between the two groups, $t = -9.69$, $p < .001$. Patients without diagnosis (10.34) had a mean WBC count approximately 3.80 units lower than patients with diagnosis (14.14).

Chi-square test conducted to check the association between Neutrophilia and Diagnosis was significant at 5% significance level (since p-value is less than 0.05). Therefore, there is evidence that there is a significant

Chi-square test for free fluids inside the abdomen and the presence of appendicitis was significant at 5% significance level, meaning that the presence of free fluids is significantly associated with the presence of appendicitis.

Cluster Analysis

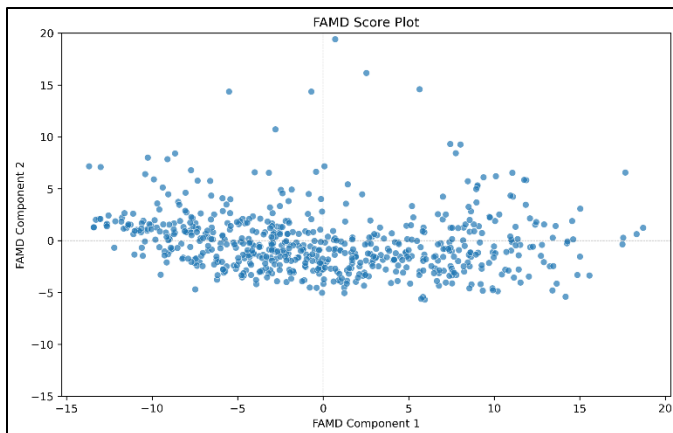


Figure 17: FAMD Score Plot with Cluster Labels

The below figure shows the FAMD (Factor Analysis of Mixed Data) plot on the mixed dataset. FAMD technique was applied to reduce the dimensionality of mixed data (both numerical and categorical variables) and visualize the data structure in two components. It was observed that there were no distinguishable or well-separated clusters in this FAMD plot. Instead, the data points of different classes were found to be overlapping and mixed with each other.

This observation is further supported by the accompanying "Elbow Method," "Silhouette Score," and "Davies-Bouldin Index" plots. For the Elbow Method, the "elbow", the point where the rate of decrease in inertia sharply slows, appears around $k = 2$, suggesting that two clusters might be appropriate. Regarding the Silhouette Score, the highest value is observed at $k = 2$, but the low score (~ 0.5) suggests poor separation between clusters. Finally, the Davies-Bouldin

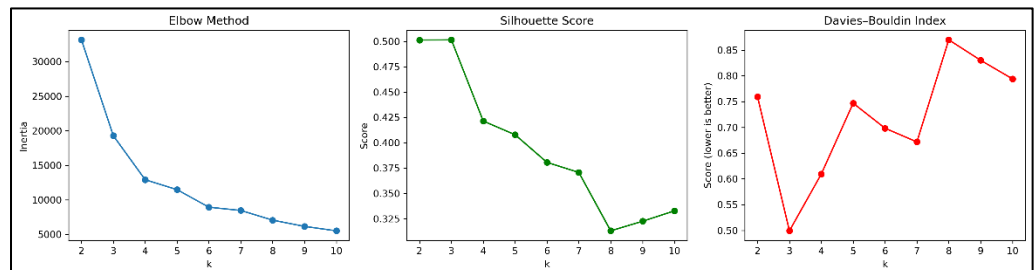


Figure 18: Optimal Cluster Determination Plots

Index shows its lowest score at $k = 2$, which numerically supports two clusters, though the separation quality is still limited.

Advanced Analysis

Model Fitting and Evaluation

After the initial preprocessing phase, the dataset was further preprocessed to train a predictive model. The numerical variables were scaled and categorical variables were one-hot encoded. Several statistical and machine learning models were evaluated using 5-fold cross-validation, and their performance was assessed based on mean cross-validation accuracy.

First, a logistic regression model with forward selection was fitted to identify the most relevant predictors for the binary outcome variable. Variables were added sequentially to the model based on their statistical significance. The process continued until no additional variables significantly improved the model. The final model obtained is shown below.

Call:

```
glm(formula = Diagnosis ~ Appendix_Diameter + Appendix_on_US +
    CRP + Peritonitis + Paedriatic_Appendicitis_Score + Free_Fluids +
    Contralateral_Rebound_Tenderness + Psoas_Sign + WBC_Count +
    Neutrophilia, family = binomial, data = train_set)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-18.655584	2.361756	-7.899	2.81e-15	***
Appendix_Diameter	2.048916	0.261299	7.841	4.46e-15	***
Appendix_on_USyes	6.281616	0.645156	9.737	< 2e-16	***
CRP	0.017880	0.004446	4.021	5.79e-05	***
Peritonitislocal	-1.342001	0.878205	-1.528	0.126483	
Peritonitisno	-3.192292	0.833272	-3.831	0.000128	***
Paedriatic_Appendicitis_Score	0.461390	0.114118	4.043	5.28e-05	***
Free_Fluidsyes	0.885602	0.334990	2.644	0.008201	**
Contralateral_Rebound_Tendernessyes	0.856353	0.346736	2.470	0.013520	*
Psoas_Signyes	-0.699854	0.355322	-1.970	0.048880	*
WBC_Count	0.083095	0.039517	2.103	0.035484	*
Neutrophiliayes	-0.836661	0.421480	-1.985	0.047138	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 841.83 on 623 degrees of freedom
 Residual deviance: 280.66 on 612 degrees of freedom
 AIC: 304.66

Table 9: Logistic regression model obtained using forward selection

Confusion Matrix and Statistics

```

Reference
Prediction 0 1
0 56 9
1 14 77

Accuracy : 0.8526
95% CI : (0.787, 0.9042)
No Information Rate : 0.5513
P-Value [Acc > NIR] : 9.726e-16

Kappa : 0.7

McNemar's Test P-Value : 0.4042

Sensitivity : 0.8953
Specificity : 0.8000
Pos Pred Value : 0.8462
Neg Pred Value : 0.8615
Prevalence : 0.5513
Detection Rate : 0.4936
Detection Prevalence : 0.5833
Balanced Accuracy : 0.8477

'Positive' Class : 1

```

Table 10: Confusion matrix of Logistic regression

After that, several machine learning models were fitted and evaluated using 5-fold cross-validation.

Model	cross-validation scores	Mean score
DecisionTreeClassifier(random_state=42)	[0.9, 0.85, 0.87, 0.88, 0.86]	0.872
RandomForestClassifier(random_state=42)	[0.91, 0.85, 0.93, 0.9, 0.89]	0.896
AdaBoostClassifier(random_state=42)	[0.93, 0.91, 0.9, 0.86, 0.9]	0.900
SVC(random_state=42)	[0.86, 0.84, 0.86, 0.87, 0.84]	0.854
KNeighborsClassifier()	[0.74, 0.78, 0.72, 0.76, 0.81]	0.762
XGBClassifier(base_score=None, booster=None, c...	[0.94, 0.9, 0.94, 0.92, 0.87]	0.914

Table 11: Performances of machine learning models

It is observed that the XGBoost Classifier gives the best accuracy score. After that, Ada Boost Classifier and Random Forest Classifier give the second and third best accuracy scores. Therefore, those top three models are used to evaluate the performance on the test set.

	precision	recall	f1-score	support
0	0.88	0.86	0.87	65
1	0.90	0.91	0.91	91
accuracy			0.89	156
macro avg	0.89	0.89	0.89	156
weighted avg	0.89	0.89	0.89	156

Table 12: Classification report of Random Forest

	precision	recall	f1-score	support
0	0.87	0.91	0.89	65
1	0.93	0.90	0.92	91
accuracy			0.90	156
macro avg	0.90	0.90	0.90	156
weighted avg	0.91	0.90	0.90	156

Table 13: Classification report of Ada Boost

	precision	recall	f1-score	support
0	0.89	0.91	0.90	65
1	0.93	0.92	0.93	91
accuracy			0.92	156
macro avg	0.91	0.92	0.91	156
weighted avg	0.92	0.92	0.92	156

Table 14: Classification report of XGBoost

After the initial model fitting and evaluation, hyperparameter tuning was carried out. After the initial model fitting and evaluation, hyperparameter tuning was carried out to optimize the model's performance. This step involved systematically searching for the best combination of hyperparameters that would improve the model's predictive accuracy. Grid Search with cross-validation was employed to explore different parameter spaces.

	precision	recall	f1-score	support
0	0.89	0.88	0.88	65
1	0.91	0.92	0.92	91
accuracy			0.90	156
macro avg	0.90	0.90	0.90	156
weighted avg	0.90	0.90	0.90	156

```
{'max_depth': None,
 'max_features': None,
 'min_samples_leaf': 2,
 'min_samples_split': 10}
```

Table 15: Tuned parameters of Random Forest

Table 16: Classification report of tuned Random Forest

	precision	recall	f1-score	support
0	0.89	0.89	0.89	65
1	0.92	0.92	0.92	91
accuracy			0.91	156
macro avg	0.91	0.91	0.91	156
weighted avg	0.91	0.91	0.91	156

```
{'learning_rate': 0.1, 'n_estimators': 200}
```

Table 17: Tuned parameters of Ada Boost

Table 18: Classification report of tuned Ada Boost

	precision	recall	f1-score	support
0	0.91	0.89	0.90	65
1	0.92	0.93	0.93	91
accuracy			0.92	156
macro avg	0.92	0.91	0.91	156
weighted avg	0.92	0.92	0.92	156

Table 20: Classification report of XGBoost

To further evaluate the performance of these three models, obtained the ROC and Precision-Recall curves for the three models and the Area Under those curves.

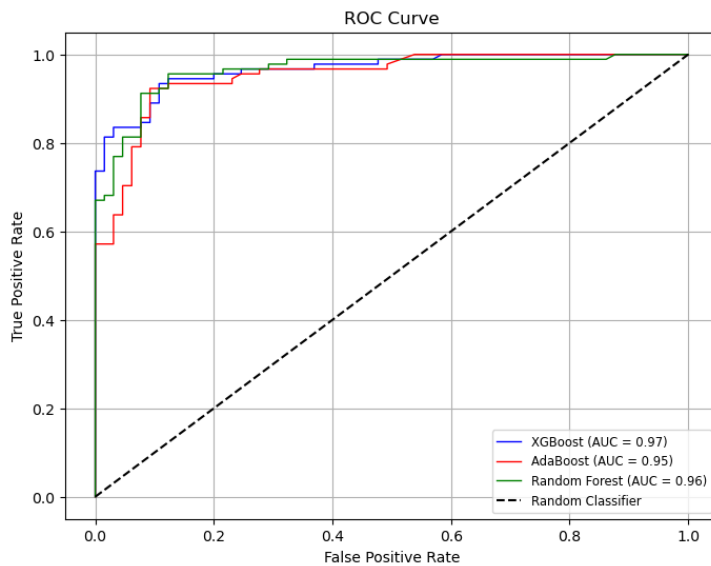


Figure 19: ROC curves

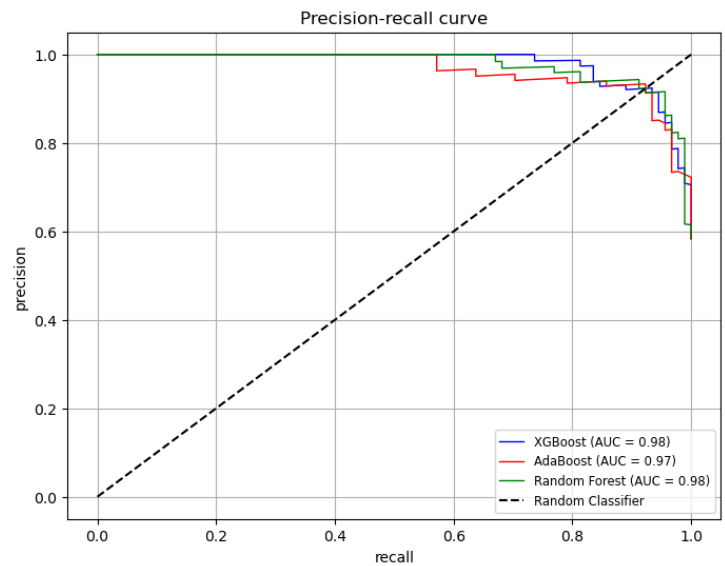


Figure 20: Precision-Recall curves

It can be observed that in both the ROC curve and the Precision-Recall curve, all three models exhibit excellent performance, but in the ROC curve, the XGBoost model gives the highest AUC value. Comparing the classification reports of the models and the above obtained curves, the **XGBoost Classifier** was selected as the **best-performing model** for the data.

After selecting the best-performing model, we examined XGBoost's built-in feature importance, which highlights how frequently and effectively features were used in tree splits. This provided an initial indication of

```
{'colsample_bytree': 0.6,
'gamma': 5,
'learning_rate': 0.2,
'max_depth': 5,
'n_estimators': 100,
'subsample': 1.0}
```

Table 19: Tuned parameters of XGBoost

which variables the model structurally relied on during training.

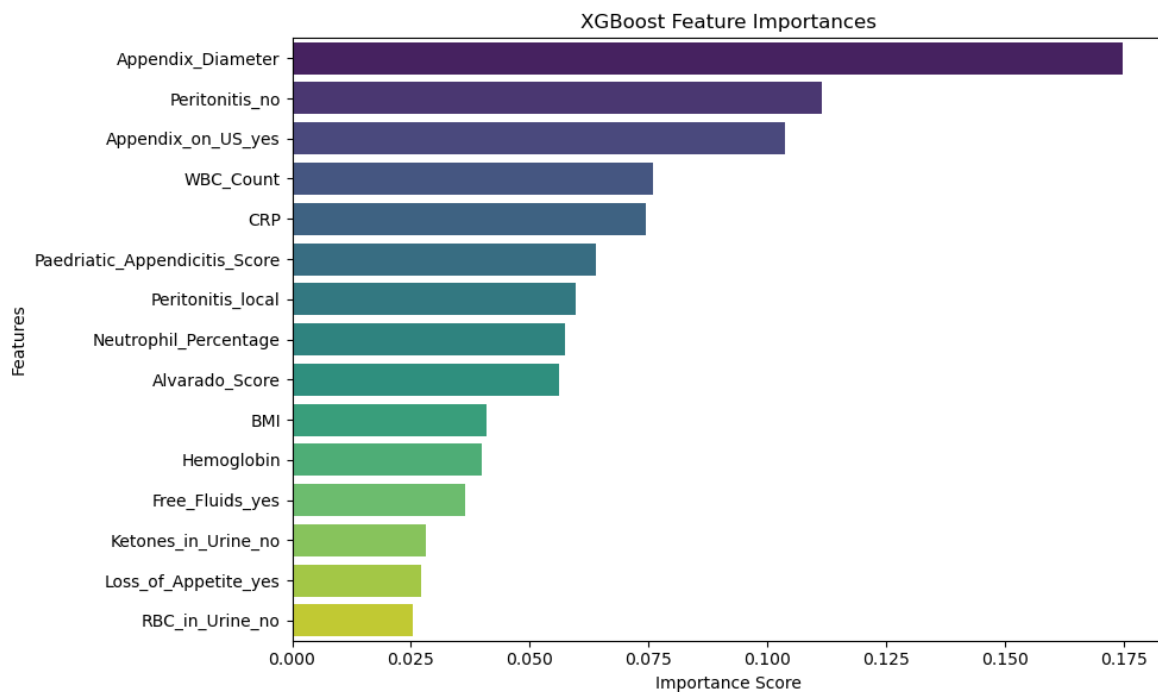


Figure 21: Feature importance plot

These are the top 15 features contributing to the predictive performance of the final XGBoost model.

After that, some of the less important features were removed from the dataset and refit the model to evaluate its performance.

Age	0.000000	Migratory_Pain_yes	0.000000
Height	0.000000	WBC_in_Urine_+++	0.000000
Weight	0.000000	WBC_in_Urine_++	0.000000
Body_Temperature	0.000000	RBC_in_Urine_+++	0.000000
RDW	0.000000	Stool_normal	0.000000
Thrombocyte_Count	0.000000	Stool_diarrhea	0.000000
Sex_male	0.000000	Stool_constipation, diarrhea	0.000000
RBC_Count	0.000000	Dysuria_yes	0.000000
Ketones_in_Urine_++	0.000000	WBC_in_Urine_no	0.000000
Ketones_in_Urine_+++	0.000000	Psoas_Sign_yes	0.000000
RBC_in_Urine_++	0.000000	Ipsilateral_Rebound_Tenderness_yes	0.000000
Neutrophilia_yes	0.000000		
Coughing_Pain_yes	0.000000		
Nausea_yes	0.000000		
Lower_Right_Abd_Pain_yes	0.000000		

Table 21: Features with less importance

	precision	recall	f1-score	support
0	0.84	0.78	0.81	65
1	0.85	0.89	0.87	91
accuracy			0.85	156
macro avg	0.84	0.84	0.84	156
weighted avg	0.85	0.85	0.85	156

Table 22: Classification report of the refitted model

Here, it is observed that the model performance is reduced compared to the full model. To increase the model performance, hyperparameter tuning was carried out on the reduced model.

	precision	recall	f1-score	support
0	0.87	0.74	0.80	65
1	0.83	0.92	0.88	91
accuracy			0.85	156
macro avg	0.85	0.83	0.84	156
weighted avg	0.85	0.85	0.84	156

```
{
  'colsample_bytree': 0.6,
  'gamma': 0,
  'learning_rate': 0.01,
  'max_depth': 5,
  'n_estimators': 300,
  'subsample': 0.6}

```

Table 23: Tuned parameters for refitted model

Table 24: Classification report of tuned refitted model

Even though hyperparameter tuning was carried out, this did not result in a performance increase in the model. It remained the same. Therefore, **the original fitted model** was selected for further analysis.

Interpret the model using Explainable AI techniques

In this section, the chosen model (XGBoost) will be interpreted using the explainable AI technique, SHAP (SHapley Additive exPlanations).

Summary Plot

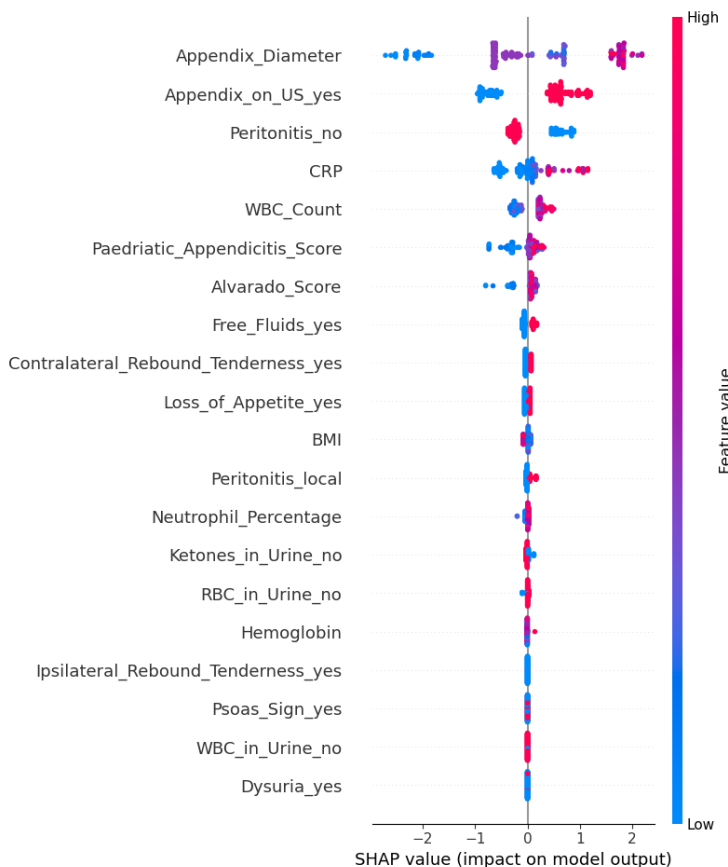


Figure 22: SHAP Summary Plot

The SHAP summary plot (Figure X) ranks features by their average contribution to the model's predictions. The results show that imaging-related variables such as **Appendix on ultrasound and Appendix diameter**, along with clinical indicators like **peritonitis status, CRP, and WBC count**, were the most influential predictors.

For example, patients with a positive ultrasound finding (red points for *Appendix_on_US_yes*) had higher SHAP values, indicating an increased probability of being classified as having appendicitis. Similarly, higher CRP and WBC counts tended to push predictions toward the positive class. In contrast, variables such as Hemoglobin, RBC in urine, and Dysuria had little impact, as reflected by their lower position in the plot.

These results suggest that the model's predictions are primarily driven by a combination of imaging findings and inflammatory markers, aligning with established clinical diagnostic pathways for appendicitis.

Dependence Plots

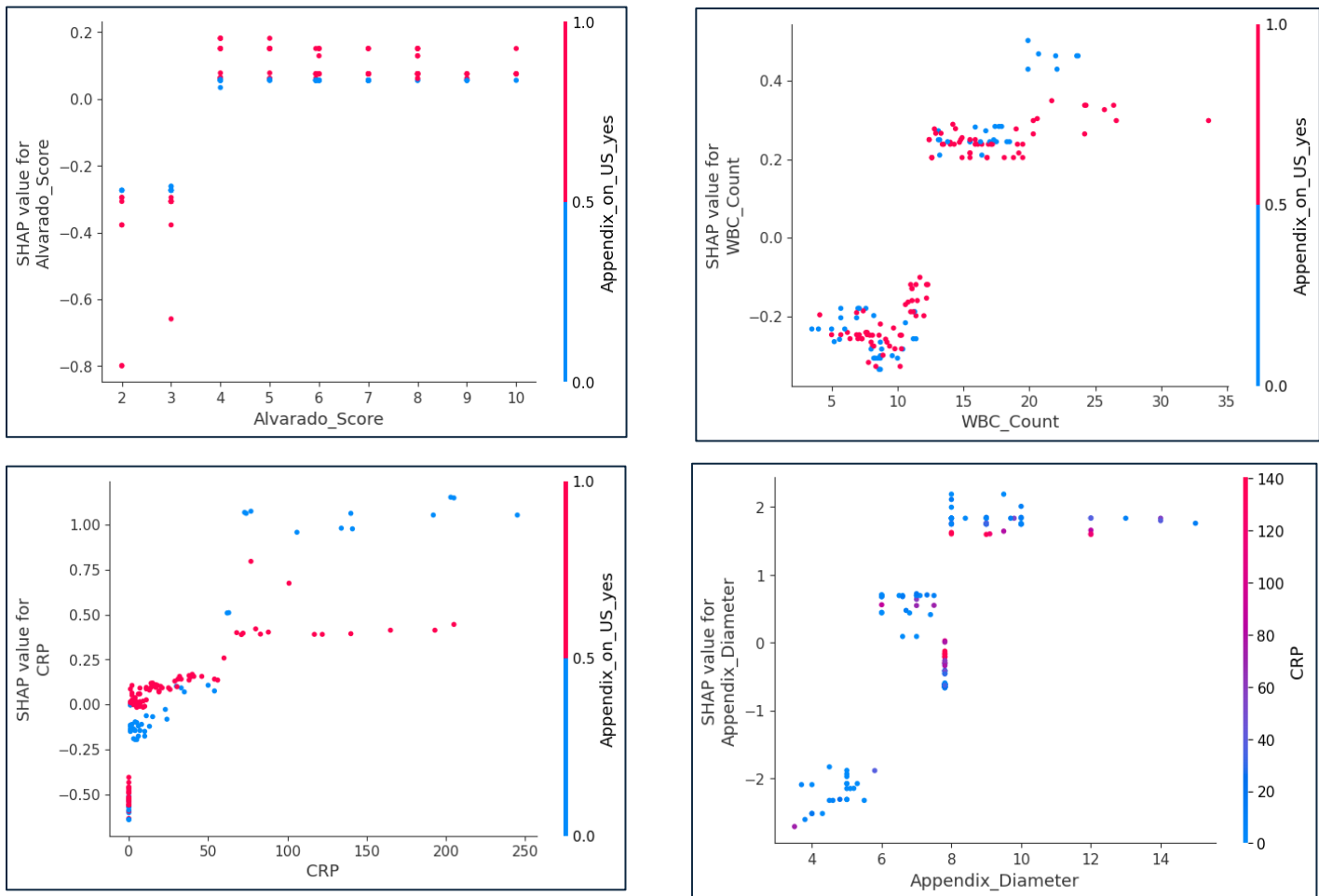


Figure 23: SHAP Dependence Plots

To further investigate how individual features influenced the predictions, SHAP dependence plots were generated. These plots reveal both the main effects of features and their interactions with other variables.

1. Alvarado_Score

The SHAP dependence plot for the Alvarado Score shows that lower values of this score are associated with negative SHAP values, meaning they reduce the predicted probability of appendicitis. As the Alvarado Score increases toward higher values, SHAP values become closer to zero or slightly positive, indicating a neutral or mild risk-increasing effect. The color gradient reflects whether the appendix was visible on ultrasound. Patients with positive ultrasound findings ($\text{Appendix_on_US_yes} = 1$, red points) tend to have higher SHAP values for the same Alvarado Score, suggesting that ultrasound findings amplify the predictive effect of the score.

2. WBC_Count

For white blood cell count, the SHAP values increase as WBC count rises. At lower WBC counts, SHAP values are negative, reducing predicted risk, but as WBC increases beyond normal ranges, the SHAP values become positive, signaling higher predicted risk.

3. CRP

C-reactive protein displays a clear upward trend: higher CRP values correspond to higher SHAP values, showing that elevated CRP is strongly associated with increased model-predicted risk of appendicitis. The interaction with appendix ultrasound findings shows mixed effect. Observations with mid SHAP values (approximately from 0 to 0.4), the appendix is visible on ultrasound (`Appendix_on_US_yes = 1`), while observations with very low and high SHAP values it is not.

4. Appendix_Diameter

The SHAP dependence plot for appendix diameter shows that smaller diameters contribute negatively to the model prediction (reduced risk), while larger diameters contribute positively, increasing predicted risk. This is consistent with the clinical understanding that appendix enlargement is a hallmark of appendicitis.

Conclusion

This study successfully demonstrated the application of machine learning techniques to improve the diagnosis of pediatric appendicitis using the Regensburg Pediatric Appendicitis dataset. Exploratory data analysis revealed important diagnostic patterns: appendicitis was more prevalent in males, right lower abdominal pain and loss of appetite emerged as strong clinical indicators, and the presence of neutrophilia and free fluids on ultrasound showed significant associations with positive diagnoses. Correlation analysis identified strong relationships between inflammatory markers (WBC and neutrophils) and clinical scoring systems (Alvarado and Pediatric Appendicitis scores), while cluster analysis using FAMD revealed substantial overlap between diagnostic groups, indicating the complexity of the classification task.

Through comprehensive data preprocessing and advanced modeling, the XGBoost Classifier emerged as the best-performing model, achieving superior performance across multiple evaluation metrics including ROC-AUC and precision-recall curves. The model's predictions were primarily driven by imaging findings (appendix visibility on ultrasound and appendix diameter) and inflammatory markers (CRP, WBC count, and neutrophil percentage), along with clinical indicators such as peritonitis status and the Alvarado Score. Using SHAP (SHapley Additive exPlanations) for model interpretation, we provided clinically meaningful insights into how individual features contribute to predictions, revealing that ultrasound findings amplify the predictive effect of clinical scores and that elevated inflammatory markers consistently increase predicted appendicitis risk—findings that align well with established clinical diagnostic pathways.

Despite attempting to create a reduced feature model, the full model maintained superior performance, suggesting that comprehensive integration of diverse medical data sources enhances diagnostic accuracy. The development of an interactive dashboard and user-friendly application translates these sophisticated machine learning insights into practical clinical tools. Overall, this study demonstrates that machine learning can effectively support clinical decision-making in pediatric appendicitis diagnosis, potentially reducing misdiagnosis rates, preventing unnecessary surgeries, and enabling faster, more accurate treatment decisions to improve patient outcomes.